

● EASY

● GEOMETRY AND TRIGONOMETRY

● ANSWER KEY

Circles

01

3 answers · Rationale-rich · Teach from doc

Q1**A****A. 9**

Choice A is correct. A circle has 360 degrees of arc. In the circle shown, O is the center of the circle and $\angle AOC$ is a central angle of the circle. From the figure, the two diameters that meet to form $\angle AOC$ are perpendicular, so the measure of $\angle AOC$ is 90° . Therefore, the length of minor arc $\overline{AC}$ is $\frac{90}{360}$ of the circumference of the circle. Since the circumference of the circle is 36, the length of minor arc $\overline{AC}$ is $\frac{90}{360} \times 36 = 9$.

Choices B, C, and D are incorrect. The perpendicular diameters divide the circumference of the circle into four equal arcs; therefore, minor arc $\overline{AC}$ is $\frac{1}{4}$ of the circumference. However, the lengths in choices B and C are, respectively, $\frac{1}{3}$ and $\frac{1}{2}$ the circumference of the circle, and the length in choice D is the length of the entire circumference. None of these lengths is $\frac{1}{4}$ the circumference.

Q2

C

C. 9

Choice C is correct. The equation of a circle in the xy -plane can be written as $x - h^2 + y - k^2 = r^2$, where the center of the circle is (h, k) and the radius of the circle is r . The graph of the given equation, $x - 6^2 + y - 3^2 = 81$, is a circle in the xy -plane. This equation can be written as $x - 6^2 + y - 3^2 = 9^2$, where $h = 6$, $k = 3$, and $r = 9$. Therefore, the radius of this circle is 9.

Choice A is incorrect. This is the y -coordinate of the center, not the radius, of the circle defined by the given equation.

Choice B is incorrect. This is the x -coordinate of the center, not the radius, of the circle defined by the given equation.

Choice D is incorrect. This is the value of the radius squared, not the radius, of the circle defined by the given equation.

Q3**C**

C. $\frac{\sqrt{3}}{2}$

Choice C is correct. The cosine of an angle is equal to the cosine of $n(2\pi)$ radians more than the angle, where n is an integer constant. Since $\frac{565\pi}{6}$ is equivalent to $47\left(2\pi\right) + \frac{\pi}{6}$, $\cos\left(\frac{565\pi}{6}\right)$ can be rewritten as $\cos\left(47\left(2\pi\right) + \frac{\pi}{6}\right)$, which is equal to $\cos\left(\frac{\pi}{6}\right)$. Therefore, the value of $\cos\left(\frac{565\pi}{6}\right)$ is equal to the value of $\cos\left(\frac{\pi}{6}\right)$, which is $\frac{\sqrt{3}}{2}$.

Alternate approach: A trigonometric ratio can be found using the unit circle, that is, a circle with radius 1 unit. The cosine of a number t is the x -coordinate of the point resulting from traveling a distance of t counterclockwise from the point $(1, 0)$ around a unit circle centered at the origin in the xy -plane. A unit circle has a circumference of 2π . It follows that since $\frac{565\pi}{6}$ is equal to $47\left(2\pi\right) + \frac{\pi}{6}$, traveling a distance of $\frac{565\pi}{6}$ counterclockwise around a unit circle means traveling around the circle completely 47 times and then another $\frac{\pi}{6}$ beyond that. That is, traveling $\frac{565\pi}{6}$ results in the same point as traveling $\frac{\pi}{6}$. Traveling $\frac{\pi}{6}$ counterclockwise from the point $(1, 0)$ around a unit circle centered at the origin in the xy -plane results in the point $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$. Thus, the value of $\cos\frac{565\pi}{6}$ is the x -coordinate of the point $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$, which is $\frac{\sqrt{3}}{2}$.

Choice A is incorrect. This is the value of $\sin\frac{565\pi}{6}$, not $\cos\frac{565\pi}{6}$.

Choice B is incorrect. This is the value of the cosine of a multiple of 2π , not $\frac{565\pi}{6}$.

Choice D is incorrect. This is the value of $\frac{1}{\tan\frac{565\pi}{6}}$, not $\cos\frac{565\pi}{6}$.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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